

A Note on More O'Farrell–Jencks Plots

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The More O'Farrell–Jencks plot¹ enables chemists to better interpret changes in transition state structure. Let us explore an example.

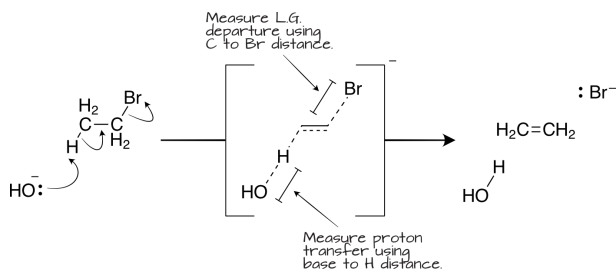
Entering the Second Dimension

Many reactions have two things happening at once in the RDS. In a typical S_N2 reaction we have the leaving group leaving and the nucleophile forming a new bond. One distance is getting longer and the other shorter as the reaction proceeds. The reaction coordinate can be plotted in three dimensions using the system popularized by Rory More O'Farrell² and William Jencks.³

These plots are especially useful⁴ in interpreting the correlation of structure with reactivity and we will be using them many more times. So let us explore a simple example using an elimination reaction.

The Example of Elimination

An elimination reaction⁵ can be described by two changes. We move a proton from the alkyl halide to a base and the leaving group departs. We could measure the former by the distance between the base and the proton; it would start at VDW distance and move toward a bond distance. We could measure the latter simply by the bond distance to the leaving group; it would change from bond distance to VDW as shown in figure 1.



One change could be complete before the other begins (the E1 mechanism) or both changes could be simultaneous (the E2 mechanism). The beauty of the More O'Farrell–Jencks plot is that we can imagine everything in-between. For example we could have a concerted E2 mechanism that has significant departure of the leaving group (and a developing positive charge) before the base makes much progress in accepting the proton. This difference in T.S. structure can be elucidated using the methods of physical organic chemistry that we will begin exploring soon. With this method we can also visualize the less common E1cb mechanism for elimination,⁵

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¹ Read ch. 7.8

² "Rory A. More O'Farrell" S.J. Ainsworth, *Chem. Eng. News*, **2012**, 90, <https://cen.acs.org/articles/90/450/RoryOFarrell.html>

³ William P. Jencks, 1927–2007" J.F. Kirsch, J.P. Richard, *Biog. Mem. Natl. Acad. Sci.*, **2010**, 3–20 <http://www.nasonline.org/publications/biographical-memoirs/memoir-pdfs/jencks-william.pdf>

⁴ For a modern example see "Mapping transition state structures for thiophosphinoyl group transfer between oxyanionic nucleophiles in water and aqueous ethanol solvents", G.I. Kalu, C.I. Ubochi, I. Onyido, *New J. Chem.*, **2022**, 46, 12981–12993. <https://doi.org/10.1039/D2NJ02008D>

⁵ Ch. 10.13

Figure 1: The E2 elimination mechanism with ethyl bromide

Consider the plots on the following page and think about the exact structure of each transition state.

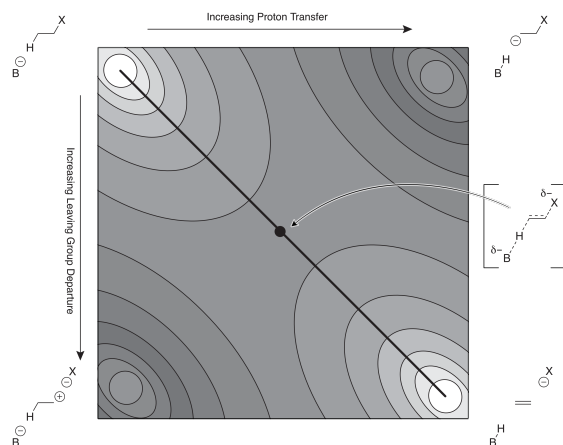


Figure 2: A More O'Ferrall-Jencks plot describing the E2 elimination mechanism

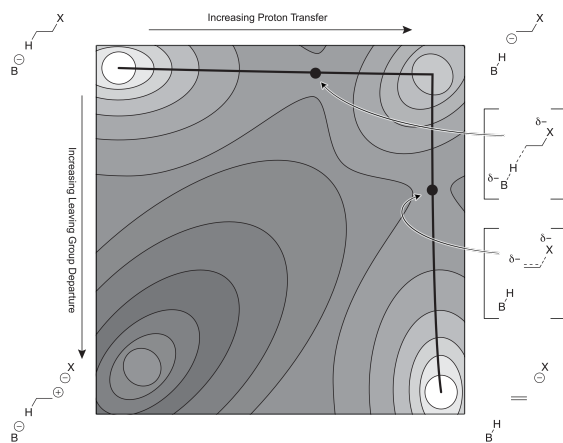


Figure 3: A More O'Ferrall-Jencks plot describing the E1cb elimination mechanism

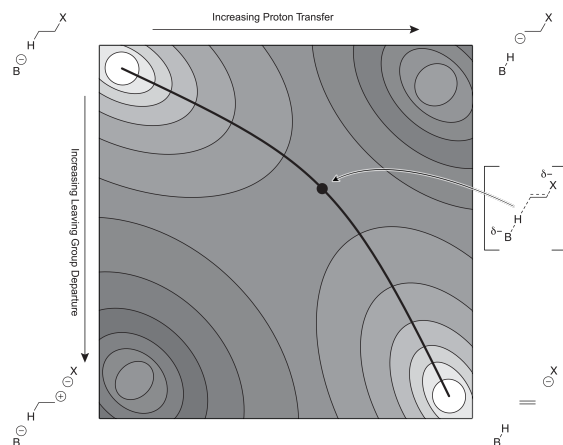


Figure 4: A More O'Ferrall-Jencks plot describing an asymmetrical E2 elimination mechanism

What path would you draw in a More O'Ferrall-Jencks plot that describes the E1 mechanism?